## GROUP 9

# LSC PROJECT

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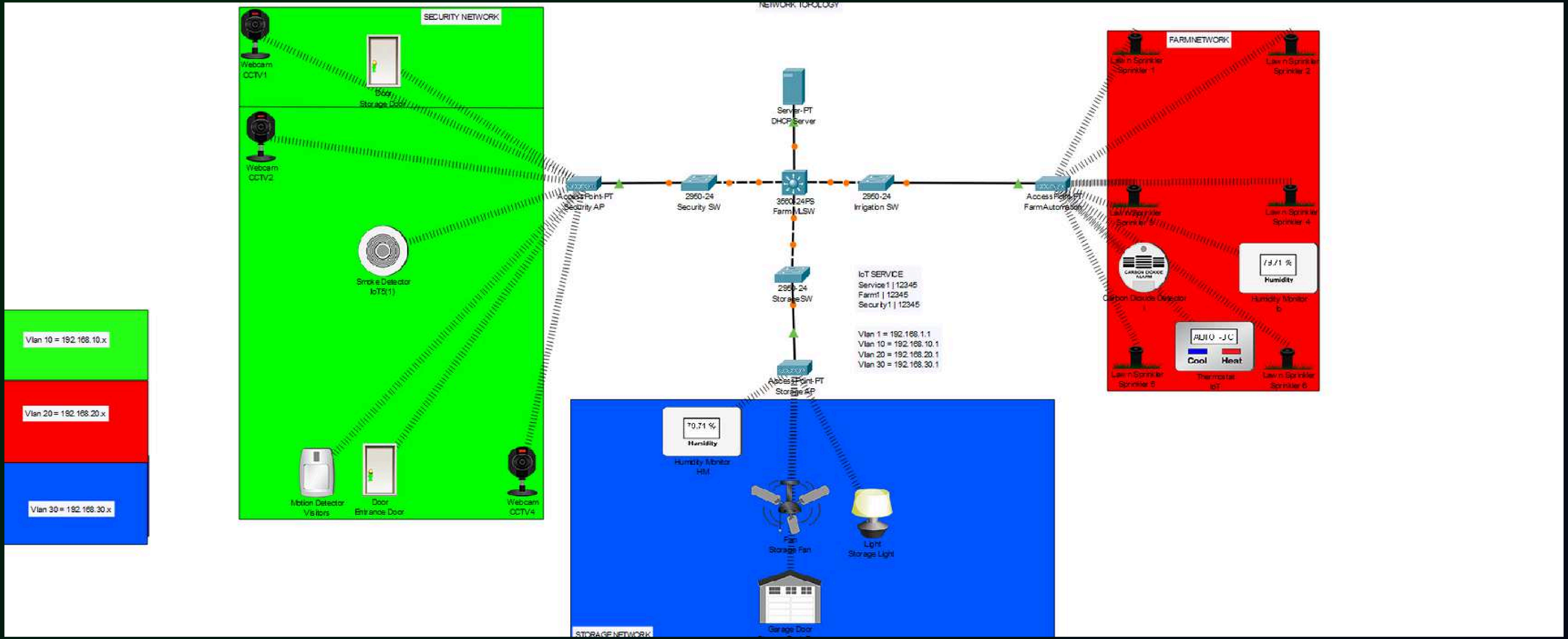
# BACKGROUND

in this project we’re continuing our previous cne project which is E-Agriculture using IoT devices. so in this project we’re mainly discuss about the server that requires for the E-Agriculture

# OUR PREVIOUS TOPOLOGY



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# SERVER THAT WE USE

01 DNS SERVER

02 DHCP SERVER

03 WEB SERVER

04 DB SERVER

03

# DNS SERVER

DNS server is a computer server that save database of ip and domain. without DNS we'll be needing to remember every ip to every domain, that's why there's DNS server to make it easier

```
options {
	listen-on port 53 { 127.0.0.1;192.168.10.2; };
	listen-on-v6 port 53 { ::1; };
	directory 	"/var/named";
	dump-file 	"/var/named/data/cache_dump.db";
	statistics-file "/var/named/data/named_stats.txt";
	memstatistics-file "/var/named/data/named_mem_stats.txt";
	secroots-file	"/var/named/data/named.secroots";
	recursing-file	"/var/named/data/named.recursing";
	allow-query     { localhost; 192.168.10.2;any;};
	allow-query-cache {localhost; 192.168.10.2;any; };
	/*
	 - If you are building an AUTHORITATIVE DNS server, do NOT enable
```

```
$TTL 86400
@	IN	SOA	dns.local. root.dns.local. (
	2025032201	;Serial
	3600		;Refresh
	1800		;Retry
	604800		;Expire
	86400		;Minimum TTL
)

@	IN	NS	ns1.dns.local.
@	IN	NS	ns2.dns.local.
@	IN	A	192.168.10.2
ns1	IN	A	192.168.10.2
ns2	IN	A	192.168.10.2
www	IN	CNAME	dns.local.
blog	IN	A	192.168.10.2
```

```
$TTL 86400
@	IN	SOA	dns.local. root.dns.local. (
	2025032201	;Serial
	3600		;Refresh
	1800		;Retry
	604800		;Expire
	86400		;Minimum TTL
)

	IN	NS	ns1.dns.local.
	IN	NS	ns2.dns.local.
@	IN	A	192.168.10.2
ns1	IN	A	192.168.10.2
ns2	IN	A	192.168.10.2
50	IN	PTR	dns.local.
50	IN	PTR	blog.dns.local.
51	IN	PTR	dhcp.local
52	IN	PTR	web.local
53	IN	PTR	db.local
```

testing:

```
[root@localhost named]# ping dns.local
PING dns.local (192.168.10.2) 56(84) bytes of data.
64 bytes from localhost.localdomain (192.168.10.2): icmp_seq=1 ttl=64 time=0.028 ms
64 bytes from localhost.localdomain (192.168.10.2): icmp_seq=2 ttl=64 time=0.088 ms
64 bytes from localhost.localdomain (192.168.10.2): icmp_seq=3 ttl=64 time=0.096 ms
^C
--- dns.local ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2041ms
rtt min/avg/max/mdev = 0.028/0.070/0.096/0.030 ms
```

# DHCP SERVER

DHCP Server is a server that's used for giving an automatic ip towards a client that connect.

Inside the dhcpd.conf:

```
subnet 192.168.10.0 netmask 255.255.255.0 {
      range 192.168.10.100 192.168.10.200;
}
```

```
[admin@localhost ~]$ systemctl status dhcpd
● dhcpd.service - DHCPv4 Server Daemon
     Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; enabled; preset: disabled)
     Active: active (running) since Sun 2025-03-23 02:39:07 WIB; 2min 59s ago
       Docs: man:dhcpd(8)
             man:dhcpd.conf(5)
   Main PID: 982 (dhcpd)
     Status: "Dispatching packets..."
      Tasks: 1 (limit: 10948)
     Memory: 6.4M
        CPU: 8ms
     CGroup: /system.slice/dhcpd.service
             └─982 /usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid
```

testing:

8. Test DHCP from Client

on the client, need to request IP from the DHCP server:

```
dhclient -v enp1s0
```

Verifying the IP:

```
ip a show enp1s0
```

output:

IoT1 (192.168.10.100)

```
[admin@localhost ~]$ ip a show enp1s0
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:b6:5d:d6 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.100/24 brd 192.168.10.255 scope global dynamic noprefixroute enp1s0
       valid_lft 34133sec preferred_lft 34133sec
    inet6 fe80::5054:ff:feb6:5dd6/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

IoT2 (192.168.10.101)

```
[admin@localhost ~]$ ip a show enp1s0
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 10
    link/ether 52:54:00:52:41:a7 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.101/24 brd 192.168.10.255 scope global dynamic noprefixroute enp1s0
       valid_lft 34173sec preferred_lft 34173sec
    inet6 fe80::5054:ff:fe52:41a7/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

# WEB SERVER

Web Server is a server that used to host a web, in this case it used to open a local web browser for our project


1. Installing

```
dnf install httpd -y
```

2. Allow httpd traffic in the firewall

```
firewall-cmd -permanent -add-service=http
firewall-cmd -reload
```

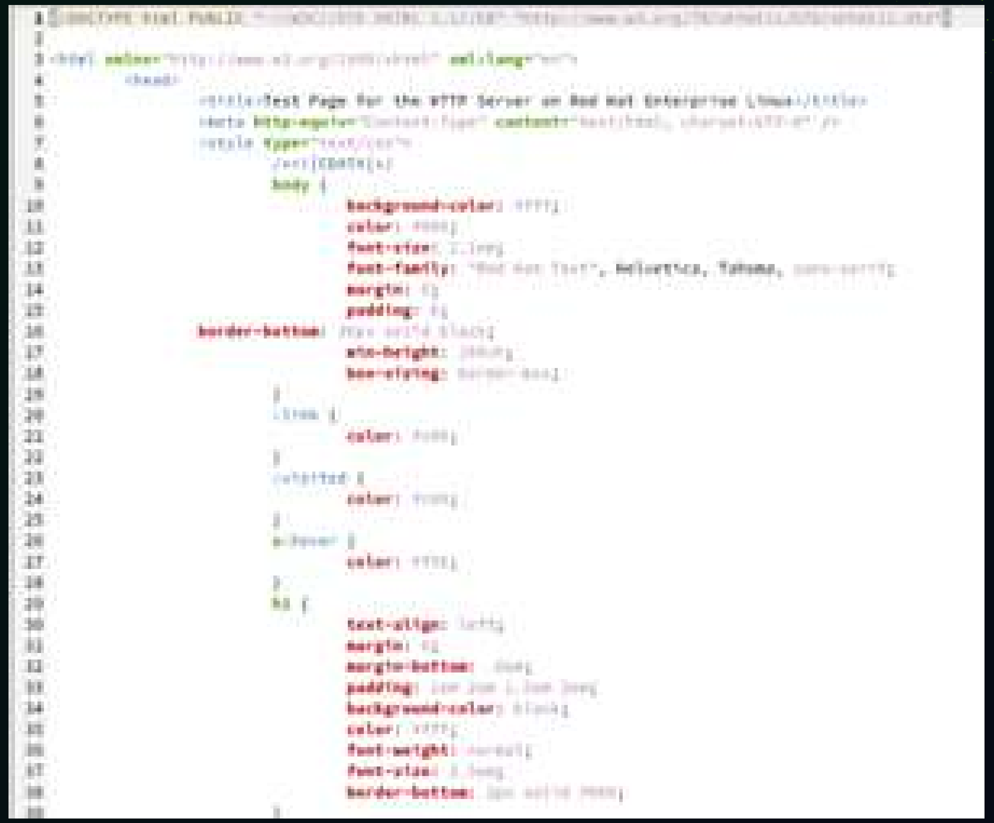
3. Assign Static IP for WEB Server

```
nmcli connection modify enp1s0 ipv4.addresses 192.168.10.4/24
ipv4.method manual
nmcli connection up enp1s0
```

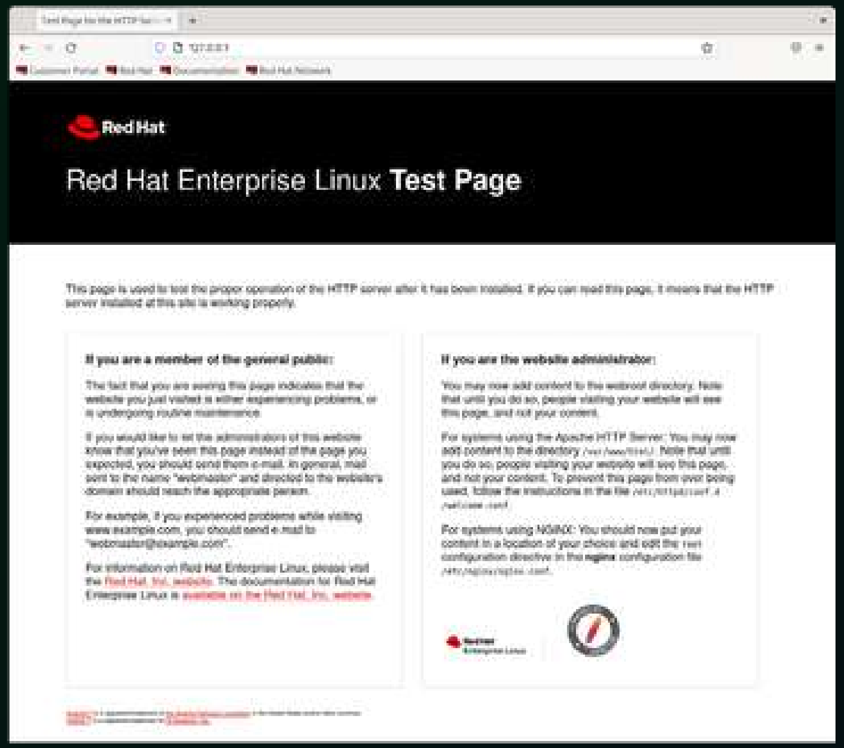
4. Start and Enable the httpd service

```
systemctl start httpd
systemctl enable httpd
systemctl status httpd
```

html.index

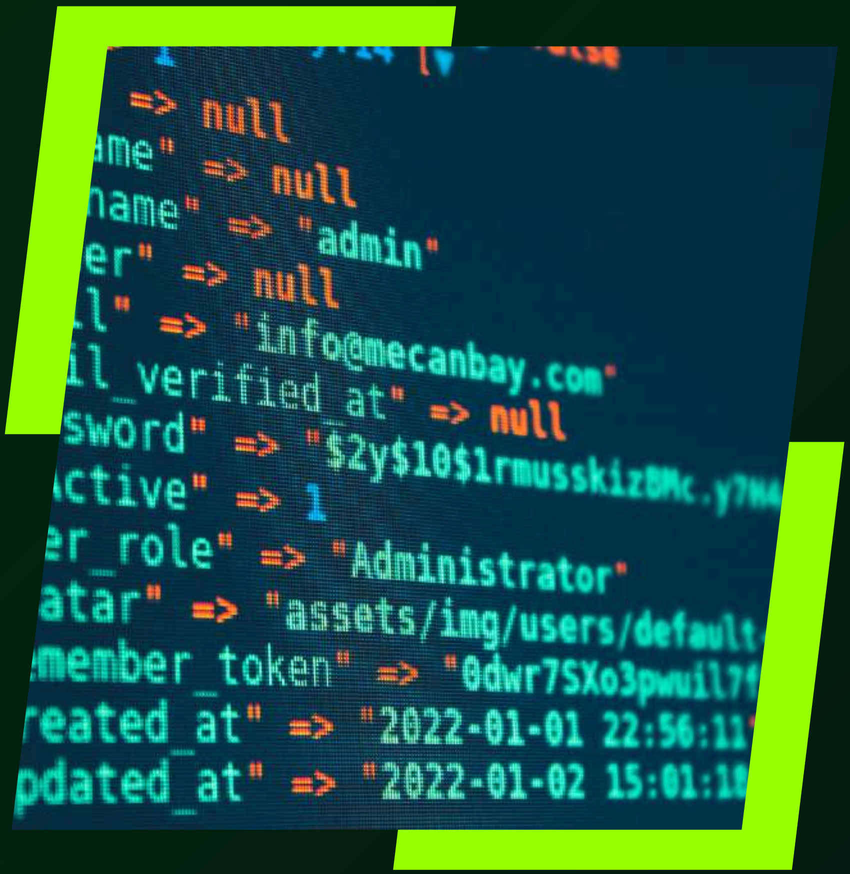


testing:



# DB SERVER

DB server or database server is a server that stored a database in a computer. this server used MariaDB as the sql language



1. Installing

```
dnf install mariadb-server -y
```

2. Start and Enable mariadb Service

```
systemctl start mariadb
systemctl enable mariadb
```

3. Check the mariadb service status

```
systemctl status mariadb
```

output:

```
[admin@localhost ~]$ systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
     Active: active (running) since Sun 2025-03-23 03:16:58 WIB; 25s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 901 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
    Process: 962 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
    Process: 1182 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
   Main PID: 1026 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 14 (limit: 10948)
     Memory: 87.0M
        CPU: 181ms
     CGroup: /system.slice/mariadb.service
             └─1026 /usr/libexec/mariadbd --basedir=/usr
```

testing:

4. Log in to mariadb

```
mysql -u root -p
```

5. Create a Database

```
CREATE DATABASE farmdb;
```

6. Create a user for remote

```
CREATE USER 'farmuser'@'%' IDENTIFIED BY 'StrongPassword123!';
GRANT ALL PRIVILEGES ON farmdb.* TO 'farmuser'@'%';
FLUSH PRIVILEGES;
EXIT;
```

# FIREWALL LIST

**01** DNS
cockpit, dhcpv6-client, dns, ssh

**02** DHCP
cockpit, dhcpv6-client, dhcp, ssh

**03** WEB
cockpit, dhcpv6-client, http, https, ssh

**04** DB
cockpit, dhcpv6-client ,ssh

# THANK YOU